RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution)

Rajalakshmi Nagar, Thandalam – 602 105

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

CR23532 – CYBER AND DIGITAL FORENSICS

ASSIGNMENT 2

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Branch	**:**	B.E. Computer Science and Engineering (Cyber Security)
Year / Semester	**:**	III Year / V Semester
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Academic Year	**:**	2026–2027

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Scenarios Selected for Each Case Study

As required in the instructions, the scenario chosen for each case study is stated below. Each one is set inside the case background given for that assignment and is used consistently in the answers.

Assignment	Case study (as given)	Scenario selected
2(1)	Suspected Unauthorized Access to a College Server	Examination question-paper files copied from the CSE department server (DEPT-SRV01) after office hours, with a USB drive recovered from a staff-room workstation
2(2)	Linking a Suspicious File to a User Using File-System Evidence	Suspicious executable, hidden files and an encrypted archive on the shared staff desktop STAFF-PC-07, linked to the account “ravi.k”
2(3)	Forensic Imaging and Safekeeping of a Suspect Storage Device	Laptop and USB drive seized from the AI research lab in a suspected research-data theft
2(4)	Recovery of Deleted Evidence from Volume Shadow Copies	Fee-concession spreadsheet altered and deleted on the accounts-office PC (ACC-PC-03), recovered from shadow copies
2(5)	Multi-Source Investigation of a Suspected Cyber Incident	Student account “stu2023_145” suspected of unauthorised access to the online lab-booking portal

Note on Approach

The answers follow the CR23532 unit content. Where a forensic tool (ILookIX, IXImager, the ASB container format) is mentioned, the answer explains the intended workflow, the evidence expected and how the results would be interpreted. It does not claim that any operation was actually carried out. All names, dates, times, serial numbers, IP addresses and file names used in the examples are hypothetical and were chosen only to demonstrate the method. Assumptions are listed at the start of each assignment, and the deliverables for each assignment are listed at its end.

ASSIGNMENT 2(1)

Suspected Unauthorized Access to a College Server

Case study: Suspected Unauthorized Access to a College Server

Scenario selected for this submission: Examination question-paper files copied from the CSE department server (DEPT-SRV01) late on a Saturday night, with a USB drive recovered from a staff-room workstation (WS-STAFF-09).

Case background (as given)

A college administrator reports that confidential examination related files were accessed and copied from a departmental server outside normal working hours. The institution has identified a small set of potential digital evidence: an administrator's complaint, a suspect workstation, server access logs, a USB device recovered from the workstation, and a few relevant files. The investigation team must preserve the evidence and determine what happened without contaminating the original evidence.

Assumptions made for this answer

- The server runs Windows Server and the shared folder holding the question papers is `\\DEPT-SRV01\ExamPapers`. File-access auditing was switched on for that folder before the incident.
- The complaint was received on Monday morning. The suspect workstation (WS-STAFF-09) is in a staff room that is unlocked on weekends, and several staff share it.
- Written authorisation from the Principal / competent authority has been obtained before any seizure or examination begins.
- All names, times and file names below are hypothetical and are used only to demonstrate the method. No tool output is claimed as actually obtained.

Question 1

Explain how the forensic team should move through the Identification, Collection, Examination, Analysis, and Presentation phases for this case.

The five phases form a fixed sequence. Each one depends on the quality of the one before it, so a shortcut early on (for example, collecting evidence without documenting it) cannot be repaired later. For this case the team would work as follows.

Phase	What the team does in this case	Output of the phase
1. Identification	Record the administrator's complaint in detail: which files, when noticed, who knew of them. Obtain written authority. Decide what could hold evidence: the server and its logs, WS-STAFF-09, the USB drive, the copied files, and non-digital sources such as the room access register and CCTV. Define the scope so that unrelated staff or student data are not touched.	Case file, authorisation letter, list of evidence sources, scene photographs
2. Collection	Secure the room and the server. Photograph everything in place. Capture volatile data first if the workstation is on (running processes, network connections, logged-in user). Export server logs with hashes, create forensic images of the workstation disk and the USB drive using a write blocker, and seal and label each item.	Forensic images, hash values, exported logs, evidence labels, chain-of-custody forms
3. Examination	Work only on verified copies. Recover the relevant data: parse the logs, list files on the workstation image, extract USB history from the registry, recover deleted files, and filter out irrelevant operating-system data so that only material related to the exam files remains.	Extracted artefacts (log entries, file lists, registry values, recovered files)
4. Analysis	Interpret the extracted artefacts. Correlate the server log entries with the workstation and USB activity, build the timeline, decide which user account was active, and test alternative explanations (for example, a shared password or a remote login).	Timeline, findings with supporting artefacts, list of unresolved points

Phase	What the team does in this case	Output of the phase
5. Presentation	Write the forensic report in plain language, attach the evidence table and timeline, and present the findings to the Principal and the disciplinary committee. Hand over the evidence with its custody record. Be ready to explain the method if the matter goes to a court or police.	Forensic report, presentation, evidence handed over with custody record

Table 1: The five forensic phases applied to the case

Question 2

Prepare a forensic investigation flowchart showing the sequence of activities from the initial complaint to presentation of findings.

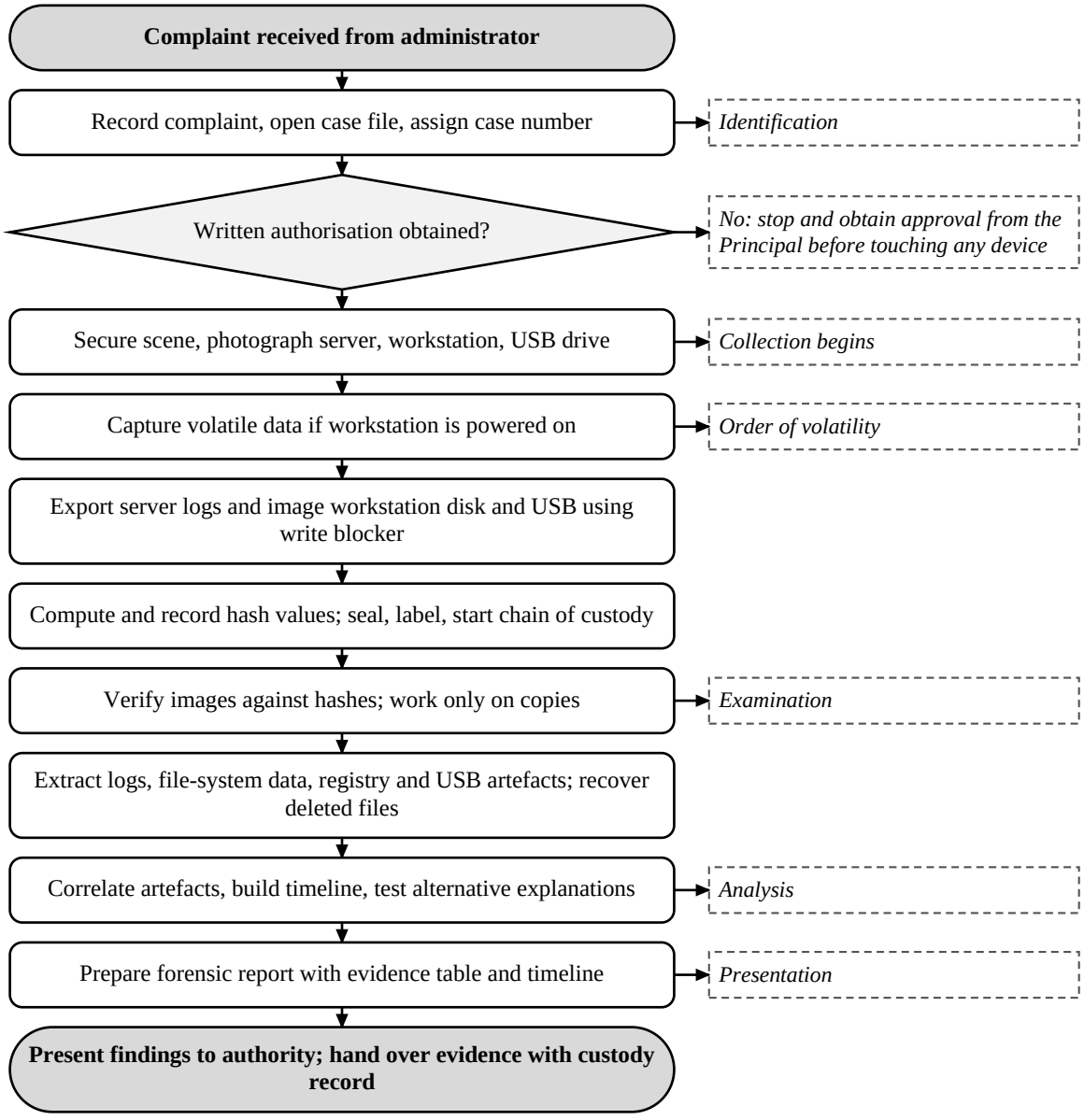


Figure 1: Investigation flowchart, from initial complaint to presentation of findings

Question 3

Identify at least five items of potential digital evidence in the case and classify the evidence according to its investigative value.

The classification below uses three levels. **High** value evidence can directly show who did what and when. **Medium** value evidence supports or corroborates the high-value items. **Low** value evidence gives background or leads only. The “type” column says whether the item is direct, corroborative or contextual.

No.	Evidence item	What it can show	Type	Value
1	Server access logs (authentication and file-access records for DEPT-SRV01)	Which account logged in, from which IP address or machine, what files were opened or copied, and the exact times	Direct	High
2	Suspect workstation WS-STAFF-09 (disk image and event logs)	User profiles, recently opened files, shortcut (LNK) and jump-list entries, USB connection history, copies of the exam files	Direct	High
3	USB device recovered from the workstation	Presence of the exam files, file timestamps, volume serial number to match the device with the workstation's USB history, deleted files	Direct	High
4	The copied exam files (on server, workstation and USB)	Metadata such as created, modified and accessed times and hash values; shows whether the copies are identical to the originals	Direct / corroborative	High
5	Administrator's complaint	Who raised the alarm, what was noticed and when; gives the initial scope and suspected time window	Contextual	Low
6	Room access register and CCTV footage	Who physically entered the staff room and the server room around the time of the activity	Corroborative	Medium
7	Network and VPN or firewall logs	Whether the login came from inside the campus or remotely; matches the IP in the server logs	Corroborative	Medium
8	Backup copies of the exam folder	The original state of the files, so that changes or deletions can be established	Corroborative	Medium

Table 2: Potential digital evidence and its investigative value

Question 4

Explain the precautions required during collection and preservation so that the evidence remains reliable and defensible.

The aim is that anyone who later examines the evidence reaches the same result, and that nobody can claim the original data was altered. The main precautions are the following.

- **Authority first.** Collect only with written permission and only within the approved scope. Evidence collected without authority may be challenged.
- **Document the scene before touching anything.** Photograph the screen, cables, connected devices and the room. Note the date and time, the clock shown by the system, and who was present.
- **Do not use the suspect computer for exploration.** Opening files, browsing folders or installing tools on the original changes access times and may overwrite data.
- **Follow the order of volatility.** If WS-STAFF-09 is switched on, record volatile information (memory, running processes, network connections) before powering down. If it is off, leave it off.
- **Use write protection.** Connect the workstation disk and the USB drive to the imaging system only through a hardware write blocker, so the original media can never be written to.
- **Image, hash, verify.** Make a bit-for-bit forensic image, calculate a hash value (for example SHA-256) of the source and of the image, and confirm that they match. Record both values.
- **Preserve server logs properly.** Export the logs in their native format, calculate hashes immediately, and keep the original log files untouched. Record how the export was made and by whom.
- **Label, seal and store safely.** Place devices in anti-static bags, seal them with signed tamper-evident tape, label with case and item number, and keep them in a locked, access-controlled store.
- **Maintain the chain of custody.** Every transfer is recorded with date, time, name, signature and purpose, so that the path of each item can be shown without gaps.
- **Work on copies only.** Examination is done on the verified image. The original stays sealed and is opened only for re-imaging if absolutely necessary, and that too is recorded.
- **Record clock differences.** Note the difference between the server clock, workstation clock and real time, so that timestamps can be compared correctly later.

Question 5

Describe how investigators can establish a timeline of the suspected unauthorized access using available evidence.

A timeline is built by collecting time-stamped events from every source, converting them to one common time reference (for example IST, noting any offset), and arranging them in order. Events that are supported by two independent sources carry more weight than those found in only one.

- **Server side:** logon and logoff events (Windows Event IDs 4624 and 4634), file-access events (4663, if object auditing was enabled), and any remote-session or share-access records.
- **Workstation side:** operating-system event logs, file-system timestamps (created, modified, accessed), recent-file and shortcut entries, and USB connection records in the registry (USBSTOR key) and the setupapi log.
- **USB drive:** the timestamps of the copied files and the volume serial number, which can be matched to the workstation's USB history.
- **Physical and network sources:** room access register, CCTV time stamps and firewall or VPN logs.

An illustrative timeline for this scenario is shown below. The times are hypothetical and are included only to show how the sources fit together.

Time (IST)	Event	Source	Weight
Sat 20:55	Person enters staff room (as recorded on CCTV and access register)	CCTV, register	Corroborative
Sat 21:03	Interactive logon to WS-STAFF-09 using account "staff09"	Workstation security log	Direct
Sat 21:10	USB drive (serial number recorded) first connected in this session	Registry USBSTOR, setupapi log	Direct
Sat 21:14	Network logon to DEPT-SRV01 from WS-STAFF-09's IP address	Server log (4624)	Direct
Sat 21:16–21:31	Multiple files in \ExamPapers opened and read	Server log (4663)	Direct
Sat 21:33	Copies of the files created on the USB drive	USB file timestamps, LNK entries	Direct
Sat 21:40	USB drive removed; logoff from server	Registry, server log (4634)	Corroborative
Mon 09:20	Administrator notices the anomaly and files complaint	Complaint	Contextual

Table 3: Illustrative timeline (hypothetical times)

Before the timeline is relied upon, the investigators must check the time-zone settings and clock accuracy of each source. If the workstation clock is, say, six minutes ahead of the server, every workstation entry must be corrected by that amount and the correction must be stated in the report.

Question 6

Explain what should be included in the final forensic report and how findings should be presented to a non-technical authority.

The report must let a reader who was not present understand what was done, why it was done, what was found, and how sure the investigators are. The sample structure below can be used.

Section	Contents
1. Cover and case details	Case number, date, investigator's name and role, the authority who requested the examination
2. Executive summary	One page in plain language: what was alleged, what was found, and the main conclusion
3. Scope and authorisation	What was asked, what was approved, and what was excluded
4. Evidence list	Each item with description, serial number, source, date and time of collection, hash values

Section	Contents
5. Methodology	Tools and versions used, write blockers, how images were made and verified, and examination steps
6. Findings and analysis	Timeline, linking of accounts, devices and files; each finding with its supporting artefact
7. Limitations and alternative explanations	Clock issues, shared accounts, data that could not be recovered, points that remain unproved
8. Conclusion	What the evidence does and does not establish. Facts are kept separate from opinions
9. Appendices	Chain-of-custody forms, screenshots, hash logs, glossary of technical terms

Table 4: Sample structure of the forensic report

For a non-technical authority such as the Principal or a disciplinary committee, the findings should be presented as follows:

- Begin with the conclusion and the level of certainty, then explain how it was reached.
- Use everyday words. Where a technical term is unavoidable, explain it in one line (for example, “a hash value is a digital fingerprint of a file”).
- Use a simple timeline chart and a small number of clear tables instead of long log extracts.
- State clearly what is a fact (for example, the account was logged in) and what is an inference (for example, the account holder was the person using it).
- Do not accuse. Report what the evidence shows and leave the decision on discipline to the competent authority.

Question 7

Briefly discuss the legal and ethical considerations involved in collecting digital evidence in a college environment.

Legal considerations. Unauthorised access to a computer system and copying of data from it can fall under Sections 43 and 66 of the Information Technology Act, 2000. For electronic records to be accepted in court, the rules on admissibility of electronic evidence must be met; under the Bharatiya Sakshya Adhiniyam, 2023 this is dealt with in Section 63 (which replaced Section 65B of the Indian Evidence Act, 1872), including the certificate that accompanies the electronic record. Since the college holds personal data of students and staff, the Digital Personal Data Protection Act, 2023 and the college’s own policies must also be kept in mind. The investigators should confirm the exact legal position with the institution’s legal adviser, and involve the police if the matter is to be pursued as a criminal complaint.

Ethical considerations.

- **Authorisation and scope:** examine only what has been approved. A college may own the equipment, but personal accounts, personal email and private messages need special care.
- **Privacy and proportionality:** access only the data needed for the investigation, and share it only with those who need to know.
- **Objectivity:** the investigator must record evidence that supports the suspect as well as evidence against, and must not begin with a fixed conclusion.
- **Confidentiality:** exam material and personal data seen during the investigation must not be disclosed or copied beyond what is required.
- **Fairness:** the suspect should be treated as innocent until the process is complete, and findings should be shared only through the proper channel.
- **Integrity of the examiner:** keep honest notes, admit mistakes, and avoid conflicts of interest (for example, an examiner who is related to or works closely with the suspect should step aside).

Deliverables covered

Investigation flowchart (Figure 1), evidence table (Table 2), case analysis (Questions 1, 4, 5 and 7) and sample forensic report structure (Table 4).

ASSIGNMENT 2(2)

Linking a Suspicious File to a User Using File-System Evidence

Case study: Linking a Suspicious File to a User Using File-System Evidence

Scenario selected for this submission: A shared staff desktop (STAFF-PC-07) in the department library room holds an unknown executable (`svc_update.exe`), hidden files, an encrypted archive and several user profiles. The account "ravi.k" is suspected of creating and opening them.

Case background (as given)

A desktop computer used by several staff members contains a suspicious executable, several hidden files, encrypted documents, browser-related artifacts, and multiple user accounts. Investigators suspect that one particular user account was involved in creating and accessing the suspicious files. The investigation team is asked to examine the hardware/software environment and use a forensic tool such as ILookIX to analyze the available evidence.

Assumptions made for this answer

- The computer runs Windows with an NTFS file system and has four local user profiles (admin, ravi.k, meena.s, guest_lab).
- Proper written authorisation exists and a forensic image of the internal drive has already been acquired and hash-verified (the acquisition procedure is covered in Assignment 2(3)).
- The ILookIX workflow is described as the intended procedure. No screenshots or tool output are claimed, and all file names, dates and values are hypothetical.

Question 1

Identify the hardware and software components that should be documented before analysis.

Documenting the environment first shows exactly what was examined, helps explain where data can reside, and protects the examiner if the results are questioned. The system inventory below should be completed before analysis starts.

Category	Items to document	Why it matters
Computer (hardware)	Make, model, serial number, asset tag, location, power state, BIOS/UEFI date and time	Identifies the exact machine; BIOS clock helps judge timestamp reliability
Storage devices	Internal HDD/SSD make, model, serial number, capacity, interface; any optical drive, card reader, USB ports in use	Each device is a possible evidence source; serial numbers tie the image to the original
Peripherals and attached media	External drives, USB sticks, printers, network cable, webcam	Shows how data could have been moved or exposed
Operating system	Name, version, build, install date, time-zone setting, last shutdown, system language	Decides which artefacts exist and where they are stored
File system and partitions	Partition layout, file-system type (NTFS), volume labels, serial numbers, free and unallocated space	Determines the metadata available and where deleted data may remain
User accounts	Account names, SIDs, groups (admin or standard), creation and last-logon dates, password policy	Needed to link activity to an account
Installed software	Browsers, office suites, encryption tools, archivers, remote-access tools, security software	Explains how encrypted files or the suspicious executable could have been made
Security configuration	Auditing settings, antivirus logs, BitLocker or other disk-encryption status	Affects what traces exist and whether the drive is readable
Forensic tools used	ILookIX and imaging tool versions, examiner name, date and time	Allows the work to be repeated and verified

Table 1: System inventory to be documented before analysis

Question 2

Explain how storage devices, operating systems, file systems, and metadata can help investigators locate relevant evidence.

- **Storage devices.** The type of drive affects what can be recovered. A traditional hard disk often keeps deleted data in unallocated space for a long time, whereas an SSD may erase it sooner because of TRIM and wear levelling. External and removable storage also needs to be identified, since the suspicious files may have been moved there.
- **Operating system.** Windows records user activity in known places: user profile folders (NTUSER.DAT, AppData), the registry, event logs, the Recycle Bin, prefetch files, and recent-file and jump-list records. Knowing the OS tells the examiner where to look for the executable's first run and the user who ran it.
- **File system.** In NTFS, the Master File Table (MFT) holds an entry for every file with its name, location, size and timestamps. Directory entries and the MFT help locate hidden files, deleted entries and renamed files. Unallocated space can hold remnants of deleted files.
- **Metadata.** Metadata is data about data: the created, modified, accessed and (in NTFS) MFT-changed times, the owner of the file, file size, attributes (hidden, system), and, for documents and images, internal properties such as author, software used and edit times. Metadata often shows who made a file and when, even if the content is unreadable.

Used together, these let the examiner narrow a very large disk down to a few relevant files. For instance, filtering for files created in a certain period by one user profile, then checking their hidden attribute and the registry entries that show they were opened, produces a short, meaningful list.

Question 3

Describe how hidden files and encrypted files may affect a forensic investigation. Discuss appropriate evidence-handling considerations.

Hidden files. A file can be hidden through the hidden or system attribute, by being placed inside system-like folders, by renaming it with a misleading extension, or by being stored in alternate data streams on NTFS. Such files are not shown in a normal folder view, so a casual inspection can miss them. A forensic tool reads the file system directly and lists them, and the examiner can compare file signatures (headers) with extensions to detect disguised files.

Encrypted files. Encryption makes the content unreadable without the key, so it can block the investigation of what the file contains. The existence, size, timestamps, name and location of the file can still be recorded and may be valuable, even when the content cannot be opened.

Evidence-handling considerations:

- Do not change the hidden attribute or open the files on the original; do all work on the forensic image and record the original attribute values.
- Document the existence and metadata of every encrypted file before any attempt to open it, and note the encryption method if it can be identified.
- Search the evidence (documents, browser data, notes, password managers) for lawful ways of obtaining the key or password, and record how the key was found. Do not guess passwords on the original device.
- If the content cannot be opened, report that fact honestly. Do not assume that an encrypted file is malicious or that it contains the data suspected.
- Check whether whole-disk encryption is active; if the system is running, the key may exist only in memory, so this has to be considered before shutdown.
- Any decrypted copy must be hashed and kept separately, with a note of how it was produced, so that it can be linked to the encrypted original.

Question 4

Prepare an evidence-linkage table showing how file names, paths, timestamps, user accounts, metadata, and other artifacts could be correlated to a user.

The table below shows how individual artefacts for this scenario could be tied to the account "ravi.k". All values are hypothetical.

Artefact	Example (hypothetical)	What it indicates	Link to user
File name and path	C:\Users\ravi.k\AppData\Local\Temp\svc_update.exe	File stored inside the profile folder of ravi.k	Moderate: location suggests the profile but other users could place files there if logged in
File owner (NTFS)	Owner: STAFF-PC-07\ravi.k	Account that created the file	Moderate to strong
Timestamps	Created 22-Sep 14:12; modified 22-Sep 14:12; last accessed 25-Sep 09:40	Creation and use times to be matched with logon sessions	Strong when matched to a logon session
User logon records	Event log: ravi.k interactive logon 22-Sep 14:05, logoff 15:30	Account was active when the file was created	Strong (supports, does not prove, the person)
Registry (NTUSER.DAT of ravi.k)	RecentDocs, UserAssist and RunMRU entries naming svc_update.exe	Executable was launched from this profile	Strong
Prefetch file	SVC_UPDATE.EXE-XXXXXXXX.pf with run count and last-run time	Program was executed and how many times	Moderate (not user-specific by itself)
Shortcut (LNK) and jump lists	LNK for secret_notes.zip in ravi.k's Recent folder	Encrypted archive was opened by this account	Strong
Browser artefacts	Download history of ravi.k's browser showing svc_update.exe saved from a web address	Source of the executable and user who downloaded it	Strong (if profile and sign-in are tied to the user)
File metadata	Archive author field or document properties showing "Ravi K"	Authorship information	Weak to moderate (can be edited)
Hidden-file attribute and ADS	Hidden attribute set on three files in ravi.k's Documents folder	Deliberate concealment or system behaviour, to be decided	Contextual

Table 2: Evidence-linkage matrix for account "ravi.k"

Question 5

Explain a suitable workflow for importing/acquiring the forensic evidence and analyzing it using ILookIX, based on the concepts covered in the syllabus.

The workflow below shows the intended procedure. It describes how the examiner would work in ILookIX and what would be recorded at each step.

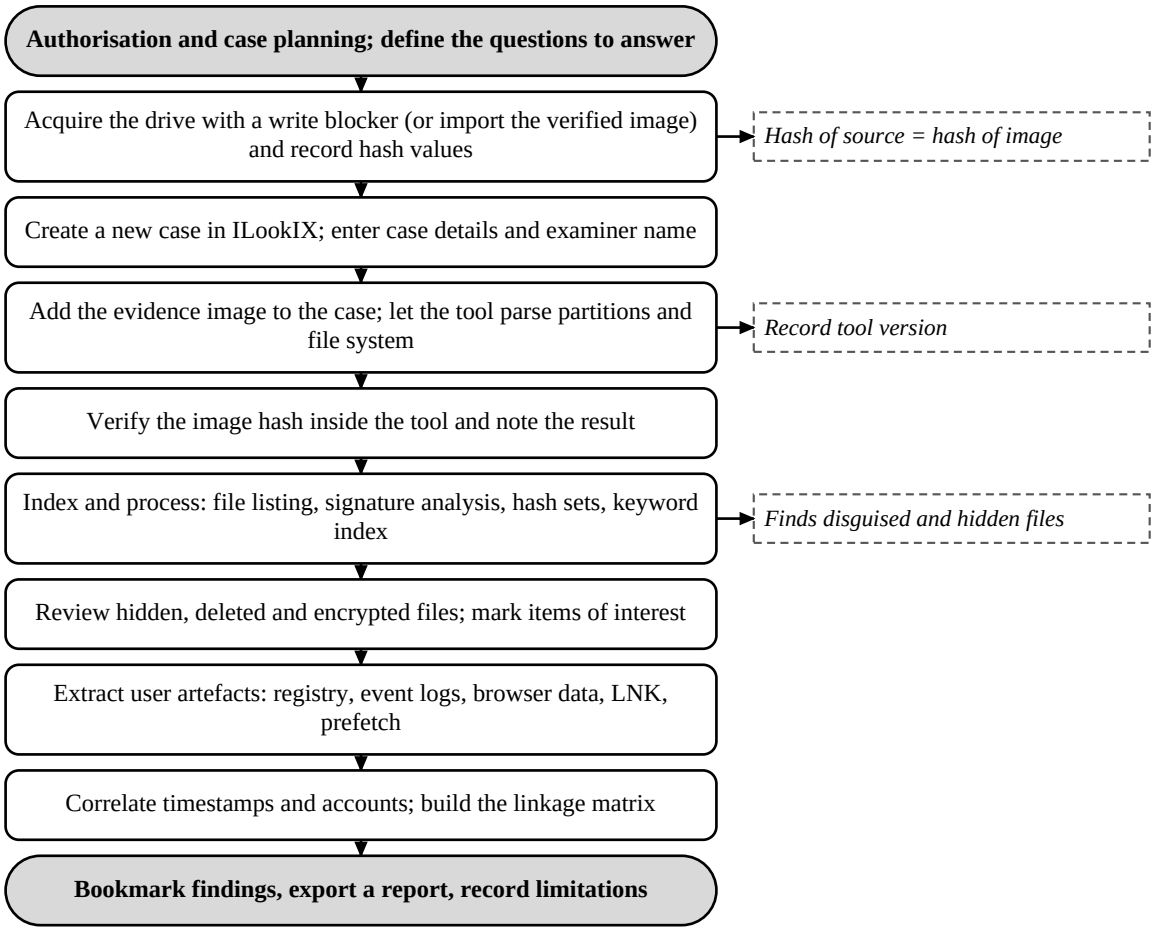


Figure 1: ILookIX analysis workflow for the suspicious-file case

Throughout the workflow, the examiner keeps case notes with the date and time of each step. The steps and the expected evidence are summarised below.

Stage	Expected procedure	Evidence expected	How it is interpreted
Import	Add the image to the case; check hash	Matching hash	Confirms the analysed data equals the acquired data
File processing	Run file listing and signature analysis	List of files with name, path, size, attributes, timestamps	Files whose extension does not match the signature are flagged for review
Hidden and encrypted items	Filter by hidden attribute and by encrypted container signatures	Hidden files, archive headers	Existence and metadata are recorded even if content stays unreadable
User artefacts	Open profile data, registry hives and logs	Logons, recent items, program execution	Links activity to an account and time
Reporting	Bookmark items and export	Report with the supporting items	Each finding can be traced to an artefact

Table 3: Stages, expected evidence and interpretation

Question 6

Differentiate between an artifact that directly links a user to an activity and an artifact that merely provides supporting/contextual information.

A **direct linking artefact** ties a specific account (and, with care, a person) to a specific action at a specific time. A **supporting or contextual artefact** shows that something happened, or helps to explain the situation, but cannot by itself show who did it.

	Direct linking artefact	Supporting / contextual artefact
Meaning	Connects an account to the act	Gives background or corroboration

	Direct linking artefact	Supporting / contextual artefact
Examples in this case	ravi.k's NTUSER.DAT showing the executable launched; LNK for the encrypted archive in ravi.k's Recent folder; browser download by ravi.k's profile; logon record overlapping the file creation time	File located in a shared folder; hidden attribute on a file; prefetch entry for the executable; presence of encryption software on the machine
Strength alone	Strong, but still tied to the account, not necessarily the person	Weak; cannot name a user
Use in the report	Core finding	Corroboration or background, with its limits stated

Table 4: Direct linking versus supporting artefacts

The strongest case is made when several direct artefacts from independent sources agree. Even then, they link an *account* to the activity; linking a *person* needs additional evidence such as who knew the password or who was physically present.

Question 7

Identify at least three limitations or alternative explanations that investigators should consider before attributing the activity to a particular user.

- **Shared or compromised credentials.** Another staff member may know the password, or the account may have been used by someone else while the owner was away. A logon only proves the account was used.
- **Malware or remote access.** The suspicious executable itself could have been planted by an attacker or installed through a download the user did not understand. Actions under an account could then be automated or remote.
- **Unreliable timestamps.** A wrong system clock, a time-zone setting, or the copying or moving of files can change created and modified times. Timestamps may also be deliberately altered with tools.
- **Shared computer.** Because several people use STAFF-PC-07, a file in a common folder or the temporary folder does not show who created it.
- **Administrator access.** A user with administrator rights can read other profiles and create files in them, so a file in ravi.k's profile may have been placed by someone else.
- **Incomplete evidence.** Overwritten data, cleared logs, undecrypted files, or limited logging may hide the real sequence of events. Absence of an artefact is not proof that something did not happen.
- **Automatic system activity.** Prefetch files, indexing and update services also create files and records, so not every artefact reflects a deliberate user action.

Conclusion with limitations

If the analysis found that ravi.k's profile owns the executable, that it was launched from that profile, that the encrypted archive's shortcut is in the same profile, and that ravi.k's logon session overlaps the file-creation time, the investigators could state that **the account ravi.k was used to create and access the suspicious files**. They should not state that ravi.k *personally* did so unless further evidence supports this (for example, CCTV or access records placing the person at the machine, or confirmation that the password was not shared). The report should also say that the contents of the encrypted files could not be examined, if that is the case, and should list the alternative explanations that were considered and how far each was ruled out.

Deliverables covered

System inventory table (Table 1), evidence-linkage matrix (Table 2), ILookIX analysis workflow (Figure 1 and Table 3) and conclusion with limitations (Question 7).

ASSIGNMENT 2(3)

Forensic Imaging and Safekeeping of a Suspect Storage Device

Case study: Forensic Imaging and Safekeeping of a Suspect Storage Device

Scenario selected for this submission: A research scholar's laptop (Item L-01, internal 512 GB drive) and an attached 64 GB USB drive (Item U-01) are seized from the college AI lab on suspicion that unpublished research data was copied out and sold.

Case background (as given)

A laptop is seized during an investigation into suspected data theft. The laptop contains an internal storage device and an attached USB drive. Investigators must collect the evidence without modifying the original media. The case requires physical acquisition, forensic imaging, preservation, data carving where necessary, and proper safekeeping. The investigation team is considering IXImager and the ASB container format for the forensic workflow.

Assumptions made for this answer

- The laptop was found powered on and locked, with the USB drive plugged in. Written authority for seizure has been issued by the competent authority.
- The imaging will be done in the college forensic laboratory using a hardware write blocker and IXImager. The procedure is described as intended; no tool output is claimed.
- Item numbers, times, serial numbers and names are hypothetical and are used only to demonstrate the procedure.

Question 1

Prepare a step-by-step procedure for safely collecting and transporting the laptop and USB device before imaging.

The goal is to bring both items to the laboratory with their data unchanged and with a clear record of what happened on the way.

1. **Confirm authority.** Check the seizure order and the scope (laptop, USB drive and accessories). Begin the case log with date, time and team members.
2. **Secure the scene.** Keep other people away from the lab bench. Do not let anyone touch the laptop, mouse or the USB drive.
3. **Photograph and sketch.** Photograph the laptop from all sides, the screen, the cable connections, the USB drive in its port, and the surroundings. Note the visible time on the screen and compare it with a reliable clock.
4. **Decide on the power state.** The laptop is on, so volatile information may be lost on shutdown. If a trained examiner is present and the policy allows, capture volatile data (memory, running processes, network connections) and note disk-encryption status before powering down. Record every action taken. If such capture is not possible, record the reason.
5. **Shut down properly.** Follow the laboratory's procedure for shutting down; do not browse, open files or plug in other devices.
6. **Remove the USB drive** only after photographing it in place, and handle it by the edges. Place it in a separate anti-static bag.
7. **Disconnect power and the battery** as per the model, so that the laptop cannot switch on or connect to a network by itself. Disable wireless if it is still enabled.
8. **Label and seal.** Assign item numbers (L-01 for the laptop, U-01 for the USB drive, L-01A for the charger). Record make, model, serial number and visible damage. Seal each bag or box with signed tamper-evident tape.
9. **Pack for transport.** Use padded, anti-static packaging. Keep items away from magnets, heat, moisture and strong vibration. Do not stack them under heavy objects.

10 **Transport under custody.** One named person carries the items directly to the laboratory, without stops, and signs the chain-of-custody form at handover. The laboratory receiver checks the seals and records their condition.

Question 2

Explain the purpose of physical acquisition and why analysis should normally be performed on a forensic image rather than the original storage media.

Physical acquisition means copying the storage device at the level of its raw sectors, from the first to the last, and not just copying the visible files. Such a copy includes the partition tables, the file-system structures, deleted files, unallocated space and slack space. This is important in a data-theft case because evidence of deleted or hidden data lives in exactly those areas.

Why work on an image and not the original:

- Every time the original is mounted or opened normally, the system can change timestamps or write data, and the evidence is altered.
- Storage media can fail. A single copy used repeatedly is a risk; with an image, the original can stay sealed.
- An image can be hashed and checked at any time. If the working copy is damaged, a fresh copy can be made from the verified image.
- Multiple examiners can work in parallel on separate copies, and an independent party can repeat the examination and obtain the same results.
- It supports the defence of the evidence in court, since the original was kept untouched.

Question 3

Design a forensic imaging workflow using IXImager. Include source identification, write protection, imaging, verification, hashing, and documentation.

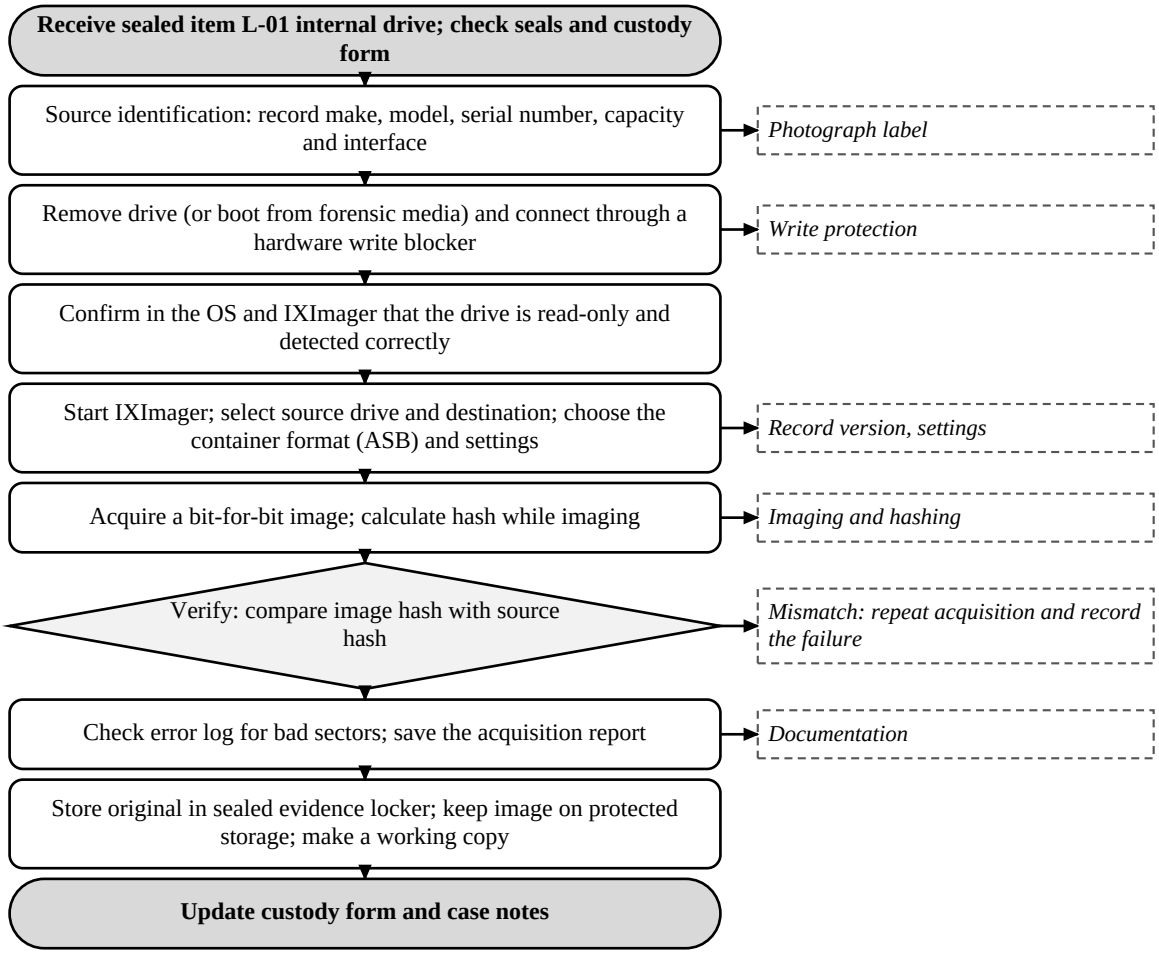


Figure 1: Imaging workflow for the internal drive using IXImager (the same steps apply to the USB drive, item U-01)

Step	Action	What is recorded
Source identification	Identify the device by make, model, serial number, size and interface; note any damage	Device details, photographs, item number
Write protection	Connect through a hardware write blocker; check that the device is read-only	Write blocker model and serial number, test result
Imaging	Use IXImager to create a full physical image into the chosen container (ASB)	Tool version, start and end time, destination path, settings
Hashing	Compute hash of the source device and of the image (for example SHA-256 as well as MD5, if required by policy)	Hash values and algorithm
Verification	Compare the hashes and check the log for read errors	Match or mismatch, any bad sectors
Documentation	Complete the acquisition record and update chain of custody	Examiner name, signature, date and time

Table 1: Imaging steps and records

Question 4

Explain how hash values can be used to demonstrate integrity before and after acquisition.

A hash function turns data of any size into a fixed-length value, so that the smallest change in the data gives a completely different hash. A hash value therefore acts as a digital fingerprint.

- **At acquisition:** the hash of the source device is calculated, and the hash of the image is calculated. If both are equal, the image is an exact copy of the source.
- **After acquisition:** whenever the image is moved, copied to a new medium or opened by another examiner, its hash is recalculated. If it still equals the recorded value, the image has not changed.
- **In the report:** the hash values are listed with the evidence, so that anybody can repeat the calculation and obtain the same value.
- **Per-segment and whole-image hashing:** where the container stores hashes per block as well as for the whole image, a damaged portion can be located without invalidating the rest.

Example (hypothetical, shortened): source SHA-256 `9F2C…B71A` and image SHA-256 `9F2C…B71A` match, so the integrity of the acquisition is demonstrated. If they differed, the image would not be used until the cause was found. Modern practice favours SHA-256 over MD5 and SHA-1, because the latter two are weak against deliberate collisions; MD5 is sometimes recorded in addition for compatibility.

Question 5

Describe the role of an ASB container in storing or managing forensic evidence and identify the information that should accompany the evidence.

In the ILookIX / IXImager environment, the ASB container is the evidence container in which the acquired data is stored. Its role is to hold the acquired image together with the details about its acquisition, so that the evidence travels as one managed unit instead of a loose file. A container of this type helps by keeping the data and the information about it together, by supporting integrity checks through stored hash values, and by being readable by the analysis tool in the same environment. (The exact internal structure should be taken from the tool documentation covered in the syllabus.)

The following information should accompany the container:

- Case number, item number and description of the source device (make, model, serial number, capacity).
- Examiner’s name, organisation, and the authority under which the acquisition was done.
- Date and time of acquisition, with the time-zone and the clock accuracy noted.
- Tool name, version, and the settings used (for example compression, segmenting, error handling).
- Hash algorithm and the source and image hash values, with the result of verification.
- Write-blocker details and any read errors or bad sectors found.
- Chain-of-custody record, storage location and the people who have accessed the container.
- Notes on encryption, passwords or special conditions of the device.

Question 6

Explain data carving and discuss when carving may be necessary if file-system information is unavailable or damaged.

Data carving is the recovery of files from raw data without depending on the file-system records. The examiner searches the sectors for known file signatures (the header and, where available, the footer of a file type; for example, a JPEG begins with the bytes FF D8 FF) and extracts the data that lies between them. Because the file name, path and timestamps are normally stored in the file system, carved files usually come back without them.

When carving becomes necessary:

- The file system is damaged, corrupted or has been formatted, so the directory entries are lost.
- Files were deleted, and their directory entries have been reused but the data blocks have not been overwritten.
- The partition table is missing or damaged and the volume cannot be mounted.
- Unallocated or slack space is examined for fragments of earlier files, for example the stolen research documents.

Limits. Carving works best when files are stored in one continuous piece. Fragmented files may be incomplete or mixed with other data; carved files lack their original names and dates, so their origin is harder to establish; and results must be validated (for example, by opening the file and hashing it) before being treated as evidence.

Question 7

Discuss date/time problems in digital evidence and explain how investigators should document time-zone, clock, and timestamp uncertainties.

Timestamps are important for the timeline, but they can mislead. The main problems are:

- **Clock error.** The computer's clock may be fast or slow, or reset after a battery failure, or have been changed by the user.
- **Time-zone and daylight-saving differences.** Some systems store time in UTC and display local time; others store local time. Different sources in the same case may use different zones.
- **Different timestamp meanings.** Created, modified, accessed and MFT-changed times behave differently. Copying a file may give the copy a new created time but keep the old modified time. Last-accessed updates may be delayed or switched off.
- **Different formats and resolutions.** FAT file systems store local time with a coarse resolution, and NTFS stores UTC in fine units, which can cause apparent differences.
- **Deliberate manipulation.** A user can alter timestamps with software or by changing the system clock.

How to document them. At seizure, the examiner compares the device clock (BIOS/UEFI and operating system) with a trusted time source and records the date, the time and the difference. The time-zone settings of the device are noted. Each timestamp quoted in the report is labelled with the zone used (for example IST, UTC+5:30) and with any correction applied, such as "device clock was 4 minutes ahead; 4 minutes subtracted". If the clock cannot be checked, the report says so and treats the timestamp as approximate. Conflicting timestamps are reported with an explanation and are not silently chosen.

Question 8

Prepare a chain-of-custody table for the seized laptop from collection through laboratory examination.

Item details: **L-01**, laptop (make and model to be entered), serial number as recorded at the scene, with item **L-01A** as the power adapter. The entries below are hypothetical and show how the form is filled in.

No.	Date and time	Released by (name, role)	Received by (name, role)	Purpose / action	Location	Condition and seal no.
1	Day 1, 10:30	Scene: Investigator A	Investigator A (collecting officer)	Seized at AI lab; photographed, powered down, sealed	AI lab	Intact; seal S-0141
2	Day 1, 11:45	Investigator A	Transport officer B	Transport to forensic lab	In transit	Seal unbroken

No.	Date and time	Released by (name, role)	Received by (name, role)	Purpose / action	Location	Condition and seal no.
3	Day 1, 12:30	Transport officer B	Lab custodian C	Handover to evidence store; seal checked	Forensic lab store	Seal S-0141 intact
4	Day 2, 09:15	Lab custodian C	Examiner D	Removal of drive and imaging with write blocker	Imaging room	Seal S-0141 broken in presence of witness; new seal S-0152 after imaging
5	Day 2, 13:00	Examiner D	Lab custodian C	Return of laptop and original drive to store	Forensic lab store	Seal S-0152 intact
6	Day 3, 10:00	Lab custodian C	Examiner D	Examination of working copy of image only (original not issued)	Analysis room	Image hash verified before use
7	Day 5, 16:30	Examiner D	Lab custodian C	Return of working media; examination complete	Forensic lab store	Sealed, seal S-0160

Table 2: Chain-of-custody record for laptop L-01 (hypothetical entries)

Each row is signed and dated by both the person releasing and the person receiving the item. Any break in the sequence, for instance a missing signature or an unexplained gap in time, must be investigated and explained in the report.

Deliverables covered

Acquisition flowchart (Figure 1), chain-of-custody form (Table 2), imaging procedure (Questions 1 and 3, Table 1) and evidence-integrity explanation (Questions 2, 4 and 5).

ASSIGNMENT 2(4)

Recovery of Deleted Evidence from Volume Shadow Copies

Case study: Recovery of Deleted Evidence from Volume Shadow Copies

Scenario selected for this submission: An accounts-office employee is suspected of altering a fee-concession spreadsheet (`Fee_Concession_List.xlsx`) and then deleting it, emptying the Recycle Bin and clearing the browser history on the office PC (ACC-PC-03).

Case background (as given)

An employee is suspected of deleting documents and browser-related artifacts after an unauthorized activity. A forensic image of the Windows system is available. Investigators discover that Volume Shadow Copies are present and believe that some deleted or changed artifacts may still be recoverable. The investigation team must select relevant evidence, remove irrelevant data, analyze artifacts, validate the findings, and reconstruct the sequence of events.

Assumptions made for this answer

- A verified forensic image of the Windows system drive of ACC-PC-03 is available, and the image contains two volume shadow copies (called SC-1 and SC-2 in this answer).
- The analysis is described as the intended forensic workflow in ILookIX. No screenshots or tool output are claimed. All dates, times and file details are hypothetical.
- Authorisation to examine the machine and the user's profile has been obtained from the competent authority.

Question 1

Explain how investigators should select and categorize digital evidence before detailed analysis.

A forensic image of a modern computer contains hundreds of thousands of files, and most are system files unrelated to the matter. Selection and categorisation reduce this to what matters and keep the analysis defensible.

- **Define the questions.** Write down what has to be answered: was the spreadsheet changed, by whom, when, was it deleted on purpose, and can it be recovered?
- **Set the scope.** Limit the time window, user profiles, and file types according to the authorisation and the questions.
- **Remove known irrelevant data.** Exclude standard operating-system and application files by comparing their hash values with a known-file hash set, and by filtering well-known system folders. Record what was excluded and why, so the filtering can be reviewed.
- **Select by relevance.** Use filters on file name, extension, path, dates and keywords (for example "fee", "concession"), and include deleted entries and unallocated space.
- **Categorise.** Place each selected item into a category (see Question 2), with the reason, so that its importance is clear.
- **Prioritise and preserve.** Examine high-value items first, bookmark them, and keep the unselected material available in case the scope changes.

Question 2

Prepare a classification scheme for user-created files, system/application files, potentially suspicious files, deleted/recovered artifacts, and supporting/contextual evidence.

Category	Description	Examples in this case	Handling
A. User-created files	Files made, saved or edited by the user	Fee_Concession_List.xlsx, related Word and PDF files in the Documents folder	Core evidence; record metadata and hash

Category	Description	Examples in this case	Handling
B. System / application files	Files created by Windows or installed programs, usually not related to the matter	Windows folder, program files, Office templates, application caches (except where relevant)	Filter out using hash sets; keep only specific items that bear on the case, such as logs
C. Potentially suspicious files	Files that look unusual in name, location, type or timing	Wiping or cleaner tools, files in the Temp folder with random names, archives, USB-copy utilities	Examine individually; note why flagged and whether it is actually suspicious
D. Deleted / recovered artefacts	Items found in the Recycle Bin, unallocated space, or recovered from shadow copies	Deleted spreadsheet from SC-1, Recycle Bin index records, recovered browser history database from SC-2	Mark as recovered; record source (volume, shadow copy ID, offset) and validate
E. Supporting / contextual evidence	Material that explains or corroborates but cannot prove the act	Event logs, USB history, recent-file shortcuts, browser downloads, user logon times	Use to build timeline and corroborate

Table 1: Evidence classification table

Question 3

Explain how Event Analysis can assist in reconstructing a sequence of activities.

Event Analysis brings together the time-stamped records from many places (file-system timestamps, event logs, registry entries, browser history, shortcut files and recovered items) and arranges them in a single chronological view. In this way, a scattered set of facts becomes a sequence of activities.

- It shows what happened before and after a key event. For example, the opening of the spreadsheet, a USB connection, the deletion and the clearing of browser history may appear within minutes of each other.
- It highlights gaps and clusters. A sudden burst of deletions followed by silence is a significant pattern.
- It allows events from different sources to be cross-checked, which increases confidence in each individual entry.
- It supports the narrative in the report, as the timeline can be shown to a non-technical reader.

Limitation: the analysis is only as reliable as the timestamps, so clock and time-zone checks must be done first.

Question 4

Explain the purpose of Cloud Analysis and Lead Analysis in an investigation and identify the kinds of questions each may help answer.

Cloud Analysis looks for traces of cloud services on the system: sync clients for cloud storage, cached or synchronised files, account identifiers, web-based email and storage activity in browser data, and links to shared files. Its purpose is to show whether evidence or stolen data may exist outside the device, and which account is used to hold it.

Lead Analysis extracts and organises leads from the evidence: names, email addresses, telephone numbers, user names, web addresses, IP addresses and keywords that appear repeatedly. Its purpose is to point the investigators to people, accounts and sources that need to be followed up, and to connect items from different parts of the evidence.

	Cloud Analysis	Lead Analysis
Purpose	Find out whether, and how, the cloud was used	Identify people, accounts and other sources to follow up
Questions it can help answer	Was the spreadsheet uploaded or synchronised to a cloud account? Which account was used? When did the sync occur? Does a copy still exist online that must be preserved through a legal request?	Whose email address or phone number appears with the file? Is the same account used on other devices? Who else received the data? What new search terms or sources should be examined?
Typical follow-up	Preservation request to the service provider, examination of the cloud account with authority	Interview of persons, examination of additional devices, further keyword searches

Table 2: Cloud Analysis and Lead Analysis compared

Question 5

Describe how Volume Shadow Copy analysis can help recover or compare earlier versions of evidence.

The Volume Shadow Copy Service in Windows creates point-in-time snapshots of a volume. They are created automatically with restore points, certain updates and backups, and are stored in the System Volume Information area. A snapshot stores the state of the volume at the time it was created, so files that were later deleted or changed can still be present in an earlier snapshot.

The analysis proceeds as follows:

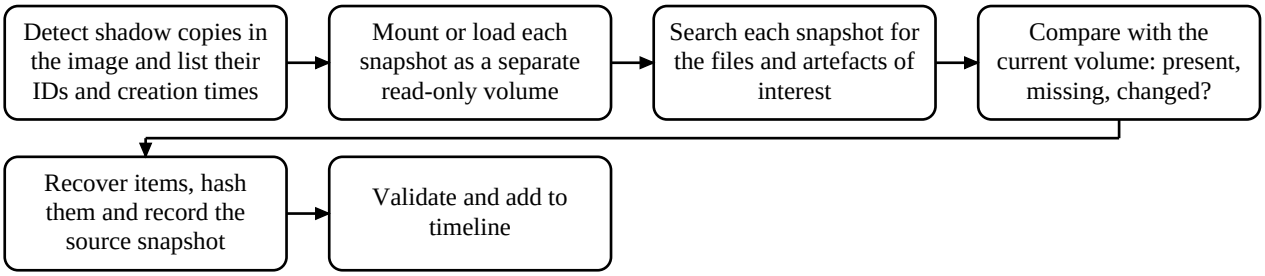


Figure 1: Volume Shadow Copy recovery workflow

- **Recovery:** a file deleted from the live volume, and not recoverable from the Recycle Bin, may still be present in a snapshot made before deletion.
- **Comparison:** the same file in different snapshots shows how it changed over time: for example, the amounts in a spreadsheet before and after modification.
- **Artefacts:** registry hives, browser history databases and logs can be recovered in their earlier state, even if they were cleared later.
- **Time bounds:** a file found in a snapshot must have existed before the snapshot's creation time, and a file missing from it did not exist, or was already deleted, at that time. This helps narrow down when events happened.

The work is done on the image, with the snapshots opened read-only, so that the original evidence is not changed.

Question 6

Construct a hypothetical timeline showing how a file could be created, modified, copied, deleted, and later recovered from a Volume Shadow Copy.

The timeline is hypothetical and meant to show how the events and the shadow copies relate. Times are in IST.

Ref.	Date and time	Event	Evidence source	Shadow copy status
T1	Mon 10 Mar, 10:05	Fee_Concession_List.xlsx created in the user's Documents folder	File system (created time)	Not yet in any snapshot
T2	Mon 10 Mar, 10:40	File modified and saved (version 1, original amounts)	File system (modified time)	-
T3	Tue 11 Mar, 02:00	Automatic restore point: shadow copy SC-1 created	Shadow copy list (creation time)	SC-1 holds version 1
T4	Tue 11 Mar, 14:20	File modified again: concession amounts changed (version 2)	File-system time, Office recent-file entries	Version 2 not in SC-1
T5	Tue 11 Mar, 14:35	File copied to a USB drive	USB history in registry, shortcut (LNK) entry, USB copy	-
T6	Tue 11 Mar, 14:50	File deleted from Documents; Recycle Bin emptied at 14:52	Recycle Bin index record, file-system change	-

Ref.	Date and time	Event	Evidence source	Shadow copy status
T7	Wed 12 Mar, 08:00	Shadow copy SC-2 created	Shadow copy list	SC-2 holds the browser history of Tuesday
T8	Wed 12 Mar, 09:00	Browser history and downloads cleared	Browser database size and time; absence of entries	Live volume has no history for Tue
T9	Wed 12 Mar, 15:00	Computer seized; forensic image made	Acquisition record	SC-1 and SC-2 in image
T10	Analysis day	SC-1 mounted: version 1 of the spreadsheet recovered. SC-2 mounted: browser history recovered. Version 1 is compared with the USB copy (version 2), showing which amounts changed	Shadow copies, USB copy, hash values	Recovered items validated

Table 3: Hypothetical event timeline with shadow copies

In this scenario the shadow copy does not contain the final version of the file, which was only on the USB drive. This is typical, since a snapshot records only what existed when it was taken. The combination of the shadow copy, the USB copy and the system records is what reconstructs the full story.

Question 7

Explain how the recovered artifact should be validated before it is included as a finding in the final report.

A recovered artefact is a candidate, not a finding, until it has been tested. The checklist below can be used.

Check	Description	Done
1. Source documented	Record the image, volume, shadow copy ID, creation time and path of the recovered item	[ ]
2. Integrity	Hash the recovered file, record the value, and confirm that it does not change when re-exported	[ ]
3. Readability	Open the file in a safe copy and confirm that it is complete and not corrupted	[ ]
4. Metadata review	Check created, modified and accessed times and the internal properties (author, last saved by)	[ ]
5. Corroboration	Compare with other artefacts: USB history, shortcut files, Recycle Bin records, event logs	[ ]
6. Second method	Repeat the recovery with a different tool or method and compare the results	[ ]
7. Time check	Verify time-zone, clock offset and shadow copy creation time	[ ]
8. Alternative explanations	Consider other users, automatic processes or software behaviour	[ ]
9. Documentation	Record tool versions, steps and examiner, so that the work can be repeated	[ ]
10. Peer review	Another examiner reviews the steps and the interpretation	[ ]

Table 4: Validation checklist for recovered artefacts

Question 8

Identify possible sources of error or alternative explanations that could affect the interpretation of timestamps and deleted artifacts.

- **Snapshot time is not event time.** A shadow copy shows that a file existed at the snapshot time. It does not show when the file was created or changed before that.
- **Clock and time-zone errors.** The system clock may have been wrong or altered, and different sources may use UTC or local time.
- **Different meanings of timestamps.** Copying or moving a file changes some timestamps but not others, so the created time of a file may be later than its modified time.
- **Deletion is not always deliberate.** Files can be removed by cleanup utilities, antivirus quarantine, application temporary-file handling, or by another user with access to the machine.

- **Browser history can be cleared automatically** by browser settings or on closing, so absence of history does not prove intentional clearing.
- **Incomplete recovery.** Snapshots can be deleted, overwritten when space is needed, or partially damaged, so missing items are not evidence that they never existed.
- **Shared use.** If several people used the account or the computer, the artefacts show the account and not necessarily the person.
- **Carved or recovered fragments** may be incomplete or mixed with other data and need careful validation.
- **Tool differences.** Different tools may present the same data differently, so results should be cross-checked.

Conclusion

On the hypothetical evidence, the reconstructed sequence shows that the spreadsheet existed in an earlier form, was changed, was copied to a USB drive, and was then deleted. The recovery from SC-1 and SC-2 and the other supporting artefacts support this order of events. Whether the act was done by the account holder and with what intention cannot be concluded from the artefacts alone, and this limit must be stated in the report.

Deliverables covered

Evidence classification table (Table 1), event timeline (Table 3), Volume Shadow Copy recovery workflow (Figure 1) and validation checklist (Table 4).

ASSIGNMENT 2(5)

Multi-Source Investigation of a Suspected Cyber Incident

Case study: Multi-Source Investigation of a Suspected Cyber Incident

Scenario selected for this submission: The student account "stu2023_145" is suspected of being used to break into the college's online lab-booking and result portal and to change a set of lab slot bookings. Evidence is spread over the student's Windows laptop, browser, personal email, a messaging app, a mobile phone and the portal's cloud-hosted records.

Case background (as given)

A student account is suspected of being used in a cyber incident involving unauthorized access to an online service. Investigators have access, subject to proper authorization, to a Windows computer, browser artifacts, email messages, messaging records, a mobile phone, and relevant Internet/cloud records. The evidence is distributed across several sources, and investigators must correlate the artifacts while considering the challenges of modern digital forensics.

Assumptions made for this answer

- Written authorisation covers the laptop, the phone, the student's college email and the portal's server-side records. Personal email and cloud accounts are accessed only through lawful process or with the student's written consent.
- All identifiers, IP addresses, times and message contents are hypothetical. Public IP addresses are taken from the documentation range (203.0.113.x) so that no real address is used. No tool output is claimed.

Question 1

Identify the potential evidence available from Windows and other operating systems in this case.

Windows (the student's laptop):

- Security and system event logs (logon and logoff, account changes, network connections, program installs).
- Registry hives: user activity (RecentDocs, UserAssist, typed paths), installed software, USB device history and network (Wi-Fi) profiles.
- Execution artefacts: prefetch files, Amcache and ShimCache entries, which show that programs such as scanners or password tools were run.
- User profile data: documents, downloads folder, desktop, email data files (OST/PST), Recycle Bin, shortcut (LNK) files and jump lists.
- Network artefacts: saved Wi-Fi names, the hosts file, DNS cache (if captured live) and any remote-access tool configurations.
- System files with leftover data: page file, hibernation file and volume shadow copies.

Other operating systems:

- **Linux** (for example a lab server or a virtual machine used by the student): authentication logs, shell command history, system and application logs, scheduled jobs, SSH keys and known-hosts files.
- **macOS** (if used): system and unified logs, property-list files with user settings, and file-activity records.
- **Mobile operating systems** (Android or iOS): app data, call and message records, location history and saved Wi-Fi networks, covered further in Question 4.
- **Server operating system of the portal:** web-server and application logs, which record each request, and are important because they show what the portal itself saw.

Question 2

Explain what browser artifacts may reveal about user activity and why browser evidence must be interpreted carefully.

What they may reveal:

- **History** of visited addresses with page titles, visit counts and times, for example visits to the portal's login page.
- **Searches** typed into search engines (for example, "how to bypass portal login").
- **Downloads** with the file name, source address and save location.
- **Cache and cookies** that hold page content and session information; they can show which sites were used and may link activity to a signed-in account.
- **Saved logins, autofill and form data** that name the accounts and email addresses used.
- **Bookmarks, session restore data and extensions**, which show interests and tools in use.

Why caution is needed:

- A visit to a page does not prove that the user wanted to be there; pop-ups, redirects, embedded content and automatic page loads also create entries.
- Several people may use the same browser profile, or one person may use several profiles and browsers, and the profile may be synchronised with another device.
- Private browsing leaves little data, and history can be cleared by the user or automatically, so missing data proves nothing.
- Time values are stored in browser-specific formats (for example, Chrome stores times as microseconds since 1 January 1601 UTC) and must be converted correctly.
- Search terms and page titles show what was typed or loaded and not why, so intent has to be shown through other evidence.
- Browser records show activity on the client side only; the portal's own logs are needed to confirm what the service received.

Question 3

List the types of evidence that may be available from email and messaging systems and explain how they can be correlated with other artifacts.

Source	Types of evidence
Email	Message body and attachments; header fields (sender, recipient, date, Message-ID, Received lines, originating IP); folders including sent, deleted and drafts; mailbox files on the laptop (OST/PST); webmail traces in the browser; login and security notices (for example, a password-reset message from the portal)
Messaging apps	Chat text, group membership, media files, time of sending and delivery, contacts, call logs, and the local app databases on the phone or laptop; backups in the cloud; deleted or disappearing messages where recoverable

Table 1: Email and messaging evidence

Correlation. Message times are compared with logon times, portal log entries and browser history. Email headers give an IP address that can be matched with the portal's access log. Attachments or files shared on chat are matched by hash with files on the laptop. Contact names, email addresses and phone numbers link the person across the email, the app and the phone. For instance, a password-reset email arriving just before a portal login, followed by a chat message such as "done, check now", builds a coherent sequence that no single source could give alone.

Question 4

Explain the forensic considerations when examining a mobile phone and identify challenges associated with mobile evidence.

Forensic considerations:

- **Authority and consent.** A phone holds very personal data, so the scope of the examination must be clear and authorised.
- **Isolate from the network.** Prevent remote wiping, new messages and sync by putting the phone in a shielded (Faraday) bag or in flight mode as per the procedure, while keeping it powered with a power bank.

- **Document the state.** Photograph the phone, record the model, IMEI, lock state, battery and the time shown, and note the SIM and memory card.
- **Choose the acquisition method.** Depending on the model and its security, the examiner may obtain a logical, file-system or physical extraction, and should use the most complete method that is possible without altering the data.
- **Hash and preserve** the extraction, work on the copy, and keep records of the tool and version.
- **Examine relevant data:** call logs, messages, app databases, browser data, photos and their metadata, location history, saved Wi-Fi networks and installed apps.

Challenges: device encryption and locked screens; a large variety of models and operating-system versions; frequent updates that change data formats; apps that store data in protected or encrypted form; disappearing messages and auto-deletion; remote wipe and cloud sync that change data after seizure; limited support of tools for new devices; and the volume and sensitivity of personal data.

Question 5

Discuss Internet and cloud-based evidence sources that may be relevant to the investigation.

- **The online service itself (portal):** web-server and application logs, authentication logs, changes in the booking database, session records, and IP addresses and user-agent strings.
- **Internet service provider and college network:** DHCP and proxy logs, firewall and Wi-Fi authentication records, which show which device and user held an IP address at a given time.
- **Email provider:** account sign-in records, message content and header information held on the server.
- **Cloud storage and backups:** files synchronised to personal cloud accounts, and phone or laptop backups that may hold deleted data.
- **Social media and messaging providers:** account and activity records, which are generally available only through lawful requests.
- **Third-party services:** hosting provider records, content delivery or DNS records, and any online tools used.

Cloud evidence is usually not held on a device in the investigator's possession, so it is obtained through the service owner (for the portal), through legal requests, or with the account holder's consent. The need to request preservation quickly is important, as retention periods vary and may be short.

Question 6

Prepare a cross-source correlation table linking at least five hypothetical artifacts from the computer, browser, email/messaging, mobile, and cloud sources.

Multi-source evidence map for the scenario:

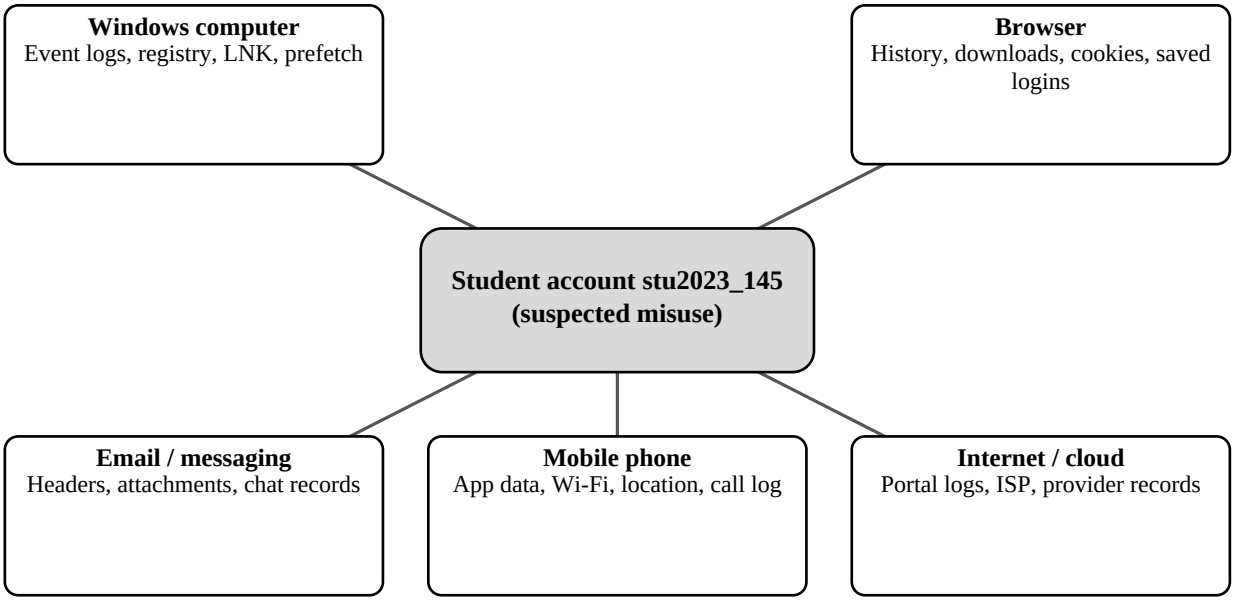


Figure 1: Multi-source evidence map

The correlation matrix below shows how hypothetical artefacts from each source can be tied together. Times are IST and hypothetical.

Ref.	Source	Artefact (hypothetical)	Time	Correlates with
A1	Computer	Windows security log: interactive logon of the student account on the laptop	Sat 22:05	Establishes the laptop was in use; matches the campus IP assignment (A6) and the portal source IP (A7)
A2	Browser	History entry for the portal's login page, followed by the booking-edit page	Sat 22:15–22:20	Matches portal login in A7 and page requests in the server log
A3	Email	Password-reset message from the portal, with the link opened	Sat 22:12	Occurs just before the login in A2 and A7; header and delivery time agree with the mail server
A4	Messaging	Chat message from the student's phone to a friend, "done, check now"	Sat 22:41	Follows the booking changes; places the phone and the student in contact with a second person
A5	Mobile	Phone connected to the college Wi-Fi network (SSID recorded) and shows location in the library	Sat 21:50	Corroborates presence on campus; matches the college network log
A6	Cloud / network	College Wi-Fi authentication log assigning IP 10.20.4.37 to the student's laptop MAC address	Sat 21:52	Links laptop to a campus IP used in A7
A7	Cloud (portal)	Portal server log: login by the student account from 10.20.4.37, then changes to 14 bookings	Sat 22:15–22:19	Matches A2, A3 and A6; gives the authoritative record of what the portal received

Table 2: Cross-source correlation matrix

The strength of the case comes from agreement between independent sources (client side and server side, device and network), and not from any single artefact.

Question 7

Explain at least six major challenges in digital forensics when evidence is distributed across multiple devices and cloud services.

No.	Challenge	Explanation
1	Volume and variety of data	Several devices and services produce a large amount of data in many different formats, making collection and review slow and requiring many tools
2	Encryption	Encrypted devices, apps and cloud data may not be readable without keys, so content may remain inaccessible
3	Jurisdiction and legal access	Cloud providers may store data in other countries, with different laws and long request procedures; access needs lawful process
4	Volatility and retention	Cloud and server logs are kept only for limited periods, and apps can auto-delete data or wipe devices remotely, so quick preservation is needed
5	Time synchronisation	Each device and service has its own clock and time-zone handling, which makes the timeline difficult to align
6	Attribution	Shared devices, shared passwords, VPNs, proxies and synchronised accounts make it hard to link an action to a person and not just an account
7	Authenticity and chain of custody	Data held by third parties is outside the investigator's control, so showing that it is complete and unaltered is harder
8	Privacy and proportionality	Personal devices and accounts hold private information not related to the case, so access must be limited and justified
9	Anti-forensics	Data wiping, timestamp alteration, private browsing and self-deleting messages can hide or distort evidence
10	Rapid technology change	New apps, operating-system versions and cloud services appear faster than tools and methods are updated

Table 3: Challenges in multi-source digital forensics

Question 8

Prepare a final investigation conclusion distinguishing direct evidence, corroborating evidence, and unresolved questions.

Summary. On the hypothetical evidence, the student account stu2023_145 was used from a campus IP address to log in to the portal and change 14 bookings during the evening of the incident. The laptop, the browser, the network records and the portal log agree on the sequence.

Category	Items	Reason
Direct evidence	Portal server log showing the login and the booking changes by the account (A7); laptop security log showing the account logon (A1); browser history of the portal pages (A2)	These show the account taking the actions, or the service recording them, at the stated times
Corroborating evidence	Password-reset email (A3); campus Wi-Fi log linking laptop to IP (A6); phone connected to campus Wi-Fi and in the library (A5); chat message “done, check now” (A4)	They support the sequence and the presence of the student at the place and time, but do not by themselves prove who made the changes
Unresolved questions	Was the student personally operating the laptop, or did someone else use it or the password? Was the password reset requested by the student or by another person with access to the email account? What was meant by the chat message? Is further evidence held by cloud or email providers that has not yet been obtained?	These need further evidence, such as CCTV, interview, provider records and comparison of typing or device data

Table 4: Direct, corroborating and unresolved findings

The final statement should therefore say that the evidence strongly supports that the account was used to access and modify the portal from the student’s laptop on campus, and that further evidence is needed before the actions can be attributed to the student personally.

Deliverables covered

Multi-source evidence map (Figure 1), correlation matrix (Table 2), challenge analysis (Table 3) and final investigative summary (Question 8, Table 4).